Rajasthan's State Health Insurance and Household Welfare

A Border-District Difference-in-Differences Study of the Bhamashah Swasthya Bima Yojana (BSBY)

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Motivation

- Indian households pay a large share of health costs **out of pocket** (OOP); one hospitalisation can push a poor family into debt.
- Policy response: **tax-funded hospital insurance**. Government pays an insurer a premium per family; families get cashless care.
  - National: RSBY (2008) $\rightarrow$ PM-JAY (2018)
  - States: Karnataka, Maharashtra, Tamil Nadu, Gujarat, ...
  - **Rajasthan: BSBY, 13 December 2015**
- Evidence is **mixed**: RSBY did not reduce OOP; Karnataka's scheme cut OOP and mortality.
- **BSBY has not been evaluated against a credible comparison group.**

Literature (1): Dube, Lester & Reich (2010)

Review of Economics and Statistics 92(4)

Introduction & objective

National studies of US minimum wages found job losses; local case studies did not. Objective: estimate effects using only **counties that touch across a state border**.

Method

County data 1990–2006 (QCEW); $\sim$316 contiguous cross-border county pairs; county FE + **pair-specific time effects**; SEs clustered by state and border segment.

Results

Earnings elasticity 0.15–0.23 (all specs). Employment: national specs -0.18 to -0.21; contiguous pairs ≈ 0 (0.016). National estimates are biased by **spatial differences in trends**.

Our use: the border-pair design, pair fixed effects, clustering.

Literature (2): Karan, Yip & Mahal (2017)

Social Science & Medicine 181

Introduction & objective

RSBY (2008): Rs 30,000 hospital cover, ∼4 crore families. Earlier studies lacked a comparison group. Objective: causal effect of RSBY on OOP spending.

Method

NSS consumer expenditure rounds 1999–2000, 2004–05 (pre) and 2011–12 (post), ∼3.5 lakh households. Poorest two quintiles in RSBY vs non-RSBY districts, propensity-score matched. DiD with **two pre-periods** (ϕ_2 pre-trend, ϕ_3 effect). States with own schemes dropped.

Results

No effect on inpatient OOP, its level, or catastrophic spending. Probability of any OOP **rose ∼30%**. Non-medical spending +5%.

Our use: comparison group's scheme; OOP benchmark; ϕ_2/ϕ_3; RSBY premium (Rs 560); Rajasthan in early RSBY rollout (p. 84).

Literature (3): Sood et al. (2014)

BMJ 349, g5114

Introduction & objective

Karnataka's Vajpayee Arogyashree (2010): free tertiary care for below-poverty-line (BPL) households, rolled out first in northern districts. Objective: effect on OOP, hospital use and mortality.

Method

Geographic regression discontinuity: 572 villages on either side of the early/late rollout boundary, propensity-matched; ∼60,000 households surveyed (Sept 2012). Placebos: above-poverty-line households, uncovered conditions.

Results

Mortality from covered conditions 0.32% vs 0.90% (−64%); OOP for covered tertiary care −64%; utilisation +44% ($p = 0.06$). No effect for ineligible households.

Our use: closest in topic and method; placebo logic; village-level RD as future work.

Literature (4): other studies

Study	Method / finding	Our use
Card & Krueger (1994)	NJ vs PA DiD; no job loss	One state vs neighbour
Abadie et al. (2010)	Synthetic control; placebo states	Permutation inference
Conley & Taber (2011)	Few treated groups: SEs too small	Why permutation
Dupas & Jain (2021)	BSBY claims; women under-served	Same scheme, other question
Kling et al. (2007)	Standardised effect sizes	SD-unit column
Dell (2010)	Spatial regression discontinuity	Future work
Dhaliwal et al. (2013)	Cost per unit of causal impact	Cost-effectiveness
Finkelstein & McKnight (2008); Hendren & Sprung-Keyser (2020)	Value of risk protection; MVPF	Future welfare work

Gap: no study estimates BSBY's effect with a comparison group.

The policy: BSBY

Design (SIHFW Rajasthan deck)

- Launched 13 Dec 2015, ∼1 crore families
- NFSA + RSBY families, ∼**67% of population**
- Rs 30,000 secondary + Rs 3 lakh tertiary per family per year
- 1,715 packages; cashless; premium paid by state
- Claims paid: **Rs 295.63 cr** (13 Dec 2015 – 10 Feb 2017)

Timeline

2005–06	NFHS-3 (pre-trend)
2008	RSBY nationally
Dec 2015	**BSBY launched**
Jan–Jul 2016	NFHS-4 in Rajasthan
2018	PM-JAY nationally
2019	BSBY $\rightarrow$ AB-MGRSBY
2019–21	NFHS-5 (post)
2023–24	NFHS-6 (third round)

Research question and hypotheses

Question

Did BSBY raise health insurance coverage and reduce out-of-pocket spending, relative to districts just across the state border that had only the national schemes?

- **H1 (coverage):** BSBY raised the share of insured households.
- **H2 (financial protection):** BSBY reduced OOP spending.
- **H3 (use of care):** BSBY changed institutional / public-facility births.
- **Placebos:** no effect on electricity or women's schooling.

Estimand: Rajasthan's scheme *relative to* RSBY/PM-JAY, not insurance vs none.

Data

Source	Use
NFHS-4 (2015–16), NFHS-5 (2019–21) district factsheets	Main analysis (one value per district)
NFHS-3/4/5 state factsheets	Pre-trend event study
NFHS-6 (2023–24), all border districts, provisional	Third round; cross-checks
Census 2011 district map	Finding border pairs
SIHFW BSBY presentation	Claims, eligibility (cost)

Unit of observation: district (Census 2011 boundaries), one value per indicator per survey round. Variables and sample: next slide.

Variables and sample

NFHS indicator (exact wording)	Unit / treatment	Role	Rounds
Households with any usual member covered under a health insurance/financing scheme	% of households	**Main outcome (H1)**	NFHS-3*, 4, 5, 6
Average out-of-pocket expenditure per delivery in a public health facility	Rs, real 2019–21 (CPI-C); capped at $Q_3+1.5$IQR; also in logs	Welfare outcome (H2)	NFHS-4, 5 (dropped in NFHS-6)
Institutional births	% of births in last 5 years	Use of care (H3)	NFHS-4, 5, 6
Institutional births in public facility	% of births in last 5 years	Use of care (H3)	NFHS-4, 5, 6
Population living in households with electricity	% of population	Placebo	NFHS-4, 5, 6
Women with 10 or more years of schooling	% of women 15–49	Placebo	NFHS-4, 5, 6

* state level only (pre-trend test).

Treatment and time

- $D_i = 1$ if the district is in Rajasthan, 0 if a neighbour
- Pre = NFHS-4 (2015–16); Post = NFHS-5 (2019–21)
- Outcome in the regression: $\Delta Y = Y^{\text{NFHS-5}} - Y^{\text{NFHS-4}}$

Sample

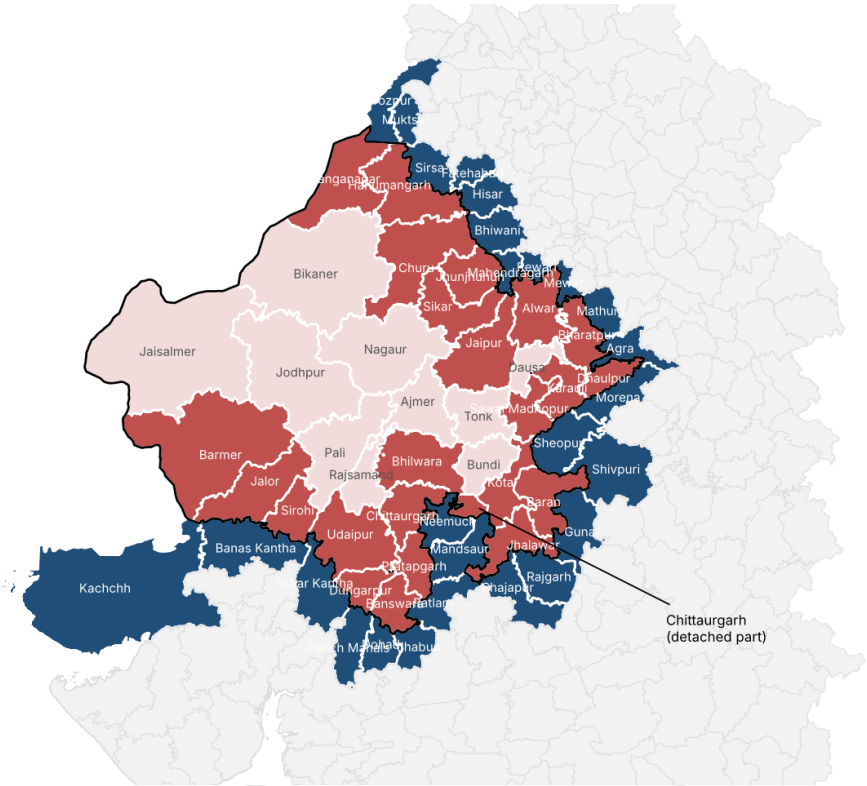
- 23 Rajasthan + 26 neighbour border districts; 51 pairs
- 5 neighbours split after NFHS-4 dropped
- **40 pairs**: 19 Rajasthan + 21 neighbour districts

Data cleaning (all checks pass)

Step	Rule / outcome
Missing values	No imputation; 0 = suppressed code $\rightarrow$ missing
Boundary changes	5 neighbour districts split after NFHS-4 dropped (51 $\rightarrow$ 40 pairs)
Cross-source	100% exact vs second copy; 33/33 vs NFHS-6 PDF
NFHS-6	Official compendiums, 6 states; 44 border districts found; 220/220 = independent compilation; Gujarat NFHS-5 insurance revision documented
Ranges & logic	% in [0,100]; public births $\leq$ institutional births: 0 violations
Inflation	OOP to real 2019–21 Rs (CPI-Combined)
Outliers	OOP capped at $Q_3 + 1.5\,\mathrm{IQR}$; **no border district reaches the cap**
Standardisation	z-scores with NFHS-4 neighbour mean and SD, both rounds
Baseline balance	Rajasthan higher OOP (+Rs 594, p = 0.09) $\Rightarrow$ baseline-control spec

Analysis scripts **refuse to run** unless the audit passes.

Border sample



23 Rajasthan × 26 neighbour districts

- Rajasthan border (treated)
- Neighbour border (control)
- Rajasthan interior (excluded)

- Districts **touching** across the state line (2011 map; ≥ 7 km shared border)
- 51 pairs: MP 22, Haryana 13, Gujarat 10, Punjab 3, UP 3
- **40 pairs** after dropping split districts
- Whole districts only

Empirical strategy

For district i in contiguous pair p:

$$\Delta Y_{ip} = \beta \cdot \mathit{Rajasthan}_i + \gamma_p + \varepsilon_{ip}, \qquad \Delta Y = Y^{\text{NFHS-5}} - Y^{\text{NFHS-4}}$$

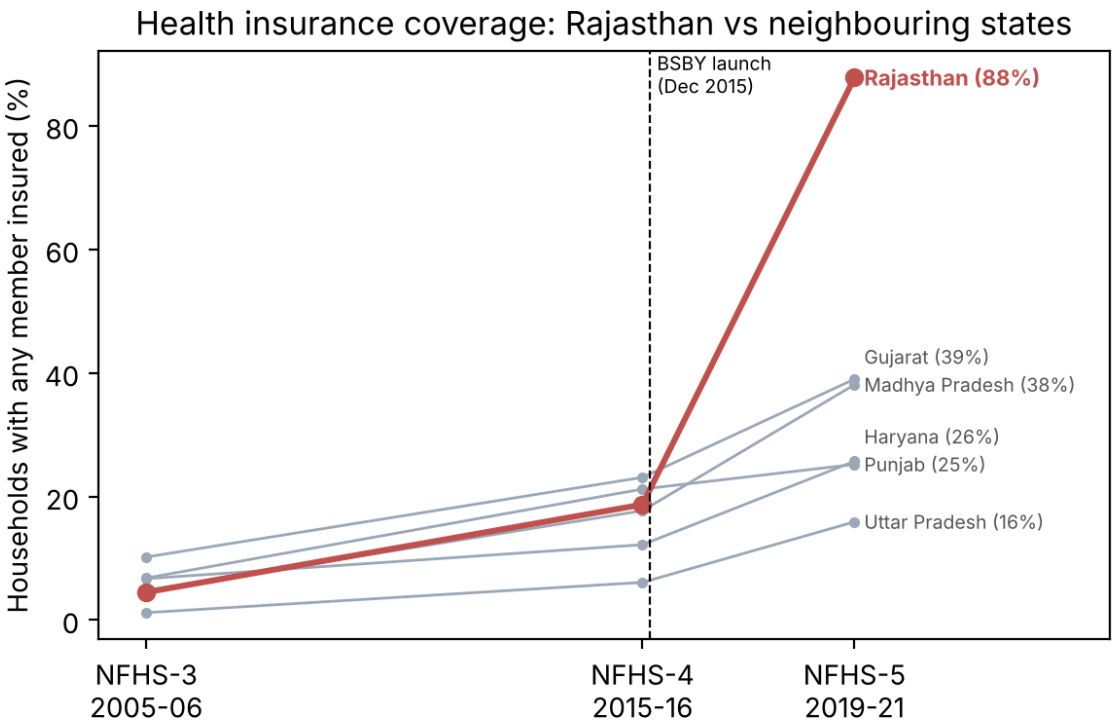
- ΔY: differencing removes everything fixed about a district (district FE)
- γ_p: pair fixed effect $\Rightarrow$ compare only with the touching neighbour
- β: within-pair difference-in-differences
- SEs **two-way clustered** by Rajasthan and neighbour district

Identifying assumption

Parallel trends across the border: without BSBY, both sides would have changed alike. Tested with a three-round state event study.

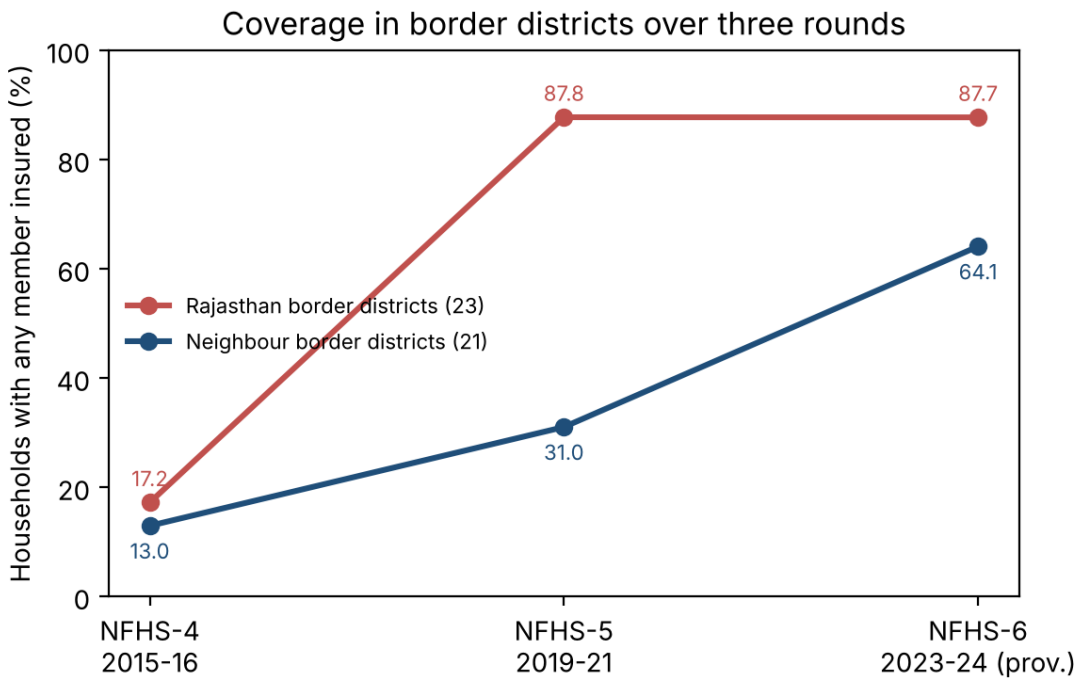
Robustness: drop Gujarat (own scheme); MP border only; pairs $\geq$ 10 km; control for NFHS-4 level; leave-one-out; OOP capping; placebos.

Pre-trends: state event study



- Pre-period DiD $\phi_2 = +4.1$ pp: roughly parallel
- Post-period DiD $\phi_3 = +56.4$ pp
- Trend-adjusted: **+52.3 pp**
- Permutation: Rajasthan **1st of 27** states, p = 0.037

Main result: coverage



NFHS-6 is provisional (official IIPS state compendiums). For 3 Gujarat districts the NFHS-6 compendium revises NFHS-5 insurance up ~5 pp.

- Rajasthan: 17.2 $\rightarrow$ 87.8 $\rightarrow$ 87.7%
- Neighbours: 13.0 $\rightarrow$ 31.0%
- Raw DiD: 70.6 $-$ 18.0 = 52.6
- **Regression: β = +52.8 pp (SE 5.1), p $<$ 0.001**
- Neighbours reach 64.1% by 2023–24 (next slide)

Did the lead last? NFHS-6 (2023–24)

Change in Rajasthan − neighbour gap	β (SE)	p
NFHS-4 → NFHS-5 (main)	+52.8 (5.1)	<0.001
NFHS-5 → NFHS-6	**−32.9 (4.7)**	<0.001
NFHS-4 → NFHS-6 (long difference)	**+15.0 (6.8)**	0.026

- Rajasthan border: 87.8 → 87.7%; neighbours: 31.0 → **64.1%** (PM-JAY)
- **BSBY's effect is mainly timing:** near-universal coverage years earlier
- Placebos flat 5→6; long-difference schooling placebo +3.6 (p = 0.04)

NFHS-6 provisional, from the official IIPS compendiums of all six states (identical to an independent compilation, 220/220 values). The Gujarat compendium revises NFHS-5 insurance for 3 districts (excluded from 4→6).

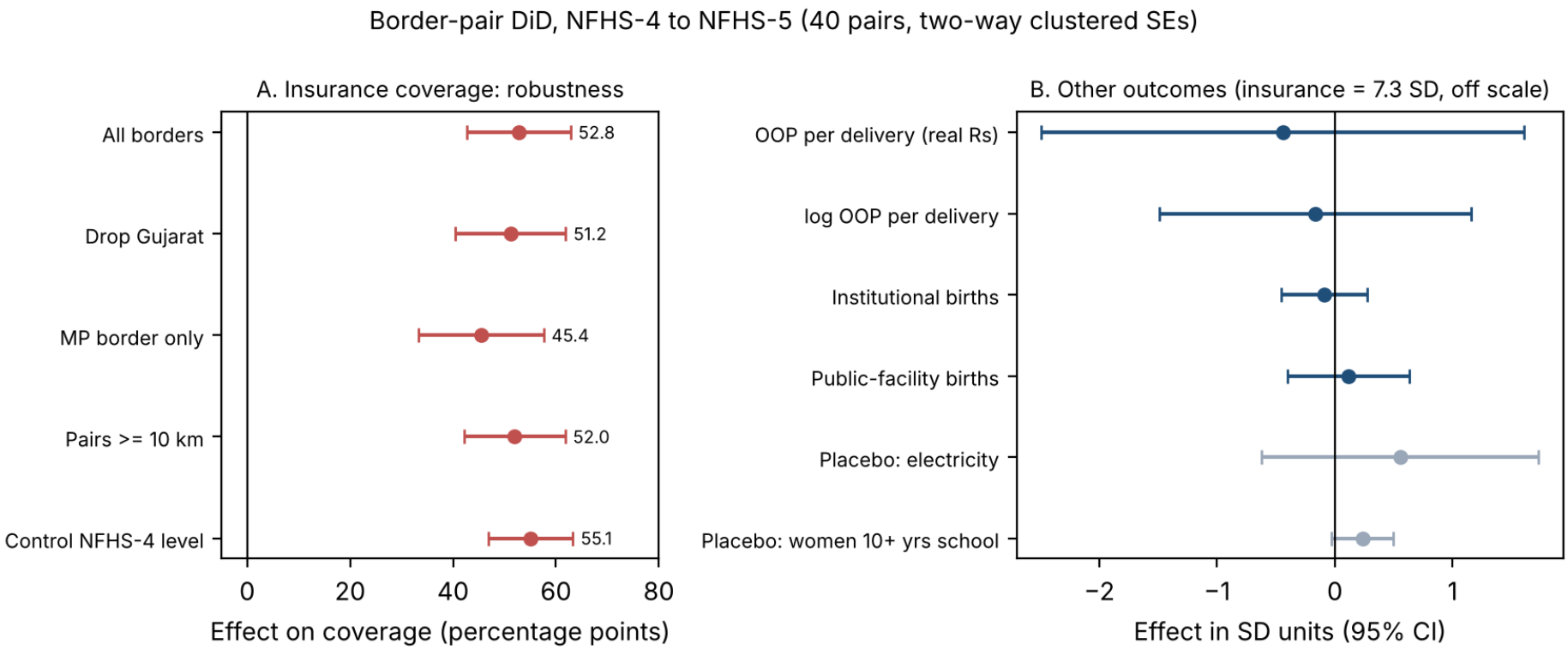
All outcomes

40 pairs, two-way clustered SEs

Outcome	β (SE)	p	SD units	+ NFHS-4 level
Insurance coverage (pp)	**+52.8 (5.1)**	**<0.001**	7.3	+55.1 (4.2)
OOP per delivery (real Rs)	−356 (840)	0.67	−0.44	+227 (212)
log OOP	−0.08 (0.31)	0.80	−0.17	+0.09 (0.12)
Institutional births (pp)	−1.2 (2.3)	0.62	−0.09	+0.8 (1.3)
Public-facility births (pp)	+2.4 (5.4)	0.66	0.12	+5.5 (3.9)
Placebo: electricity (pp)	+3.2 (3.5)	0.35	0.56	−0.1 (0.6)
Placebo: women 10+ yrs school	+3.1 (1.7)	0.08	0.24	+2.8 (1.5)

- Coverage: drop Gujarat +51.2; MP only +45.4; pairs $\geq$10 km +52.0; leave-one-district-out +51.3 to +53.9
- Nothing else significant in any sample; women's-schooling placebo borderline

Robustness at a glance



Panel A: coverage across specifications. Panel B: other outcomes in SD units (95% CI); grey = placebos.

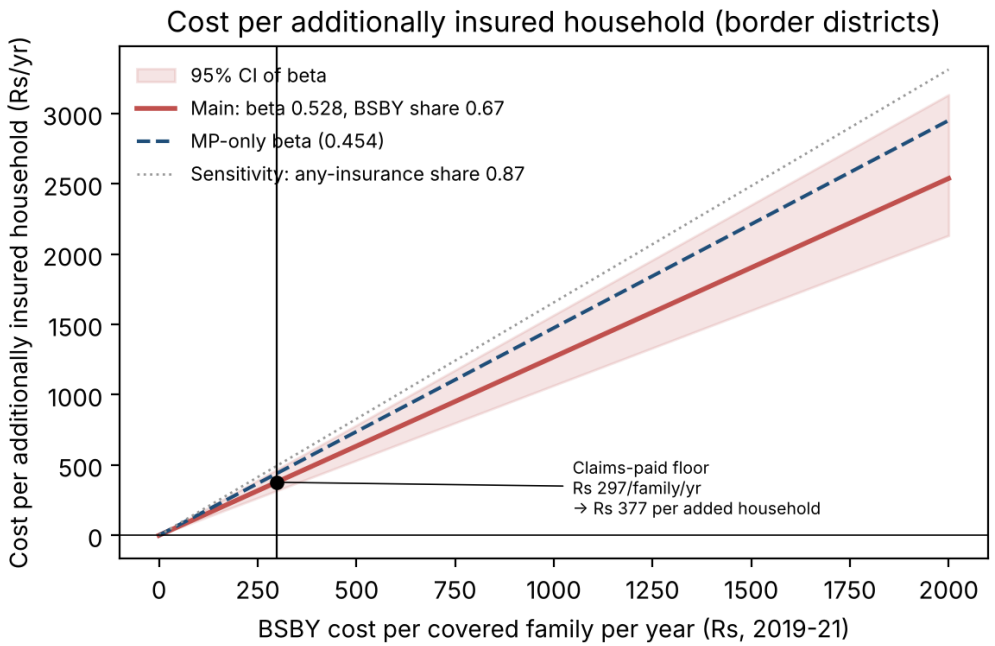
Why no effect on OOP?

Underpowered, not a zero

95% CI: −Rs 2,003 to +Rs 1,291 vs baseline Rs 2,235.
Minimum detectable effect = 2.8 × SE = **Rs 2,352** > **the whole baseline**.

- **Mean reversion:** Rajasthan started higher (+Rs 594); controlling for the starting level flips the sign to +Rs 227
- **Fragile:** dropping one district (Baran) moves it to +Rs 30
- **Wrong margin:** public-facility deliveries free under JSSK since 2011, on both sides; BSBY's value is in hospitalisations
- **Consistent with Karan et al. (2017):** RSBY did not reduce OOP

Cost-effectiveness



$$\frac{\text{cost/family} \times s_{BSBY} - p_{RSBY}\, c_{cf}}{\beta}$$

- Cost/family = **Rs 297**/yr: claims paid (Rs 295.63 cr, 1.16 yrs, 1 crore families), a *floor* on the premium
- $s_{BSBY} = 0.67$ (SIHFW p. 17)
- $p_{RSBY} = 0$ (early rollout, Karan p. 84)

Result (incremental cost-effectiveness ratio; Dhaliwal et al. 2013)

At least **Rs 377** per additionally insured household per year (CI of β: Rs 317–466; MP-only: Rs 439), measured against the 2019–21 gap. **Break-even:** each such household must gain $\geq$ Rs 377/yr. As neighbours catch up, the cost per unit of *lasting* advantage rises.

Discussion

- **Coverage:** border estimate (+52.8) and trend-adjusted state estimate (+52.3) agree; by 2023–24 PM-JAY narrowed the gap to $\sim$15 pp, so BSBY's gain was mainly *timing*.
- **Financial protection:** our OOP null agrees with RSBY evidence (Karan et al.) and does not contradict Karnataka's tertiary-care gains (Sood et al.): we measure a margin insurance barely touches, at district resolution.
- **Who benefits:** Dupas & Jain find women use BSBY less; household coverage cannot see this.
- **Value for money:** $\geq$ Rs 377 per additionally insured household; $\sim$40% of spending reaches households that had some insurance anyway.

Conclusions

1. **BSBY worked as a coverage policy, and worked early:** +53 pp relative to the border by 2019–21, robust to every check; neighbours caught up to within 15 pp by 2023–24.
2. **Financial protection is unproven, not disproven:** OOP per delivery is the wrong margin and the design is underpowered for it.
3. **Cost:** at least Rs 377 per additionally insured household per year (against the 2019–21 gap); benefits must exceed this to justify it.
4. **Policy:** a state scheme can deliver near-universal coverage years before the national scheme does; evaluations should measure hospitalisation spending and health, not just enrolment.

Limitations and future work

Limitations

- NFHS-4 baseline began 6 weeks after launch $\Rightarrow \beta$ is a lower bound
- Scheme vs scheme, not insurance vs none
- District averages; large districts
- OOP: deliveries only, underpowered
- One pre-period, state level
- Cost is a claims floor; benefits not monetised
- NFHS-6 provisional; Gujarat NFHS-5 insurance revised
- Long-difference schooling placebo significant (p = 0.04)

Future work

- NSS 71st vs 75th: OOP per hospitalisation
- NFHS GPS clusters: spatial RD (Dell 2010; Sood et al. 2014)
- Link neighbours' catch-up to PM-JAY rollout
- Actual premium $\Rightarrow$ MVPF / value of risk protection

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Thank you.

Code, data and full explainer:
`https://github.com/ipsanmoljj/india-welfare-analysis`